

CS360 Introduction to Database

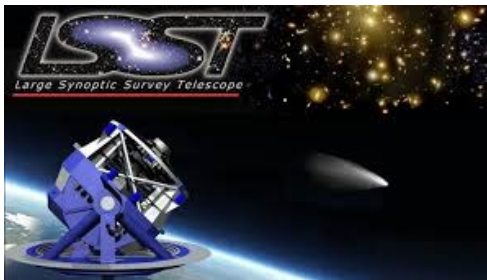
The Worlds of Database Systems

Min-Soo Kim

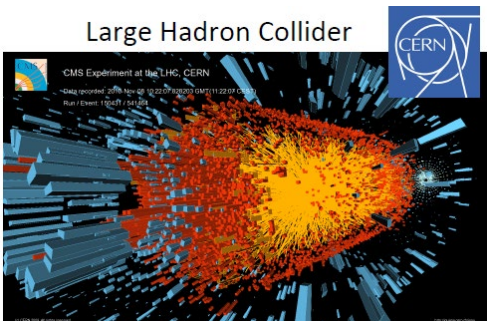
School of Computing, KAIST

The worlds of databases

- Databases today are crucial for **every business**
 - e.g., Google, Microsoft, Amazon, Tesla, ...
 - companies maintain all their important records in databases
- Databases are also used in **many scientific research**
 - e.g., data gathered by astronomers, data generated by large hadron collider, human genome, ...



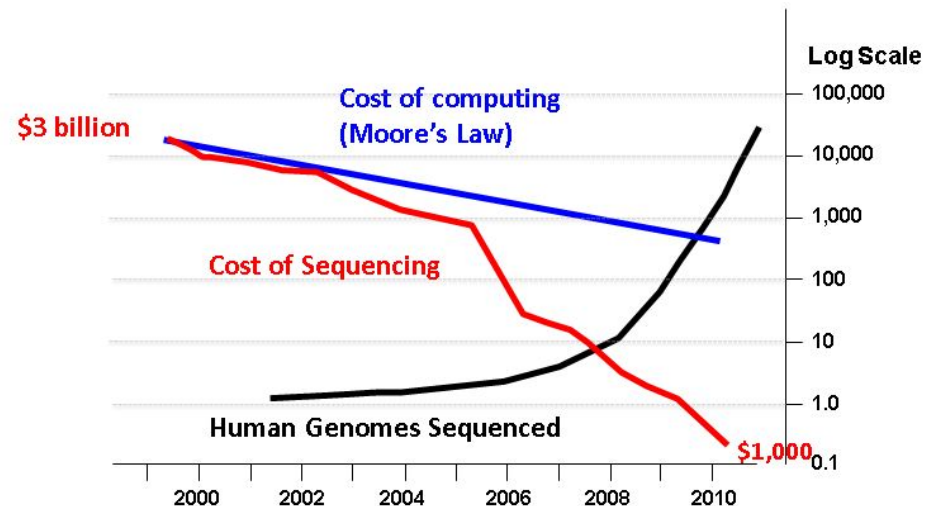
LSST
(30 TB / day)



Large Hadron Collider
(petabyte scale)

Adapted from
The Economist

The Sequencing Explosion



Database management system (DBMS)

- Database: a collection of data that is managed by a DBMS
 - power of databases comes from DBMS
- DBMS (more casually a “database system”) (c.f., OS)
 - one of the most complex software (> 10,000,000 lines)
 - creating and managing large amounts of data **efficiently**
 - making data to **persist** over long periods of time, **safely**

ORACLE®
DATABASE


Microsoft®
SQL Server®


MySQL®


PostgreSQL


mongoDB

DBMS is expected to

- Allow users to create new DBs and specify their **schemas**
 - schema: logical structure of the data
- Give users the ability to **query** data and modify data
 - query: database lingo for a question about the data
 - e.g., Structured Query Language (SQL) for relational DBMS
- Support the storage of **very large amounts of data**, allowing **efficient access** to data for queries and modifications
- Enable **durability**, the recovery of a DB from failures
- **Control access** to data from many users at once

Traditional applications of DBMS (OLTP)

- Banking systems
 - maintaining accounts
 - assuring that system failures do not cause money to disappear
- Airline reservation systems
 - assuring that data will not be lost
 - accept very large volumes of small actions by customers
- Corporate record keeping
 - employment and tax records
 - inventories, sales records
 - ...

Modern applications of DBMS

- Business intelligence (BI)
 - OnLine Analytical Processing (OLAP) for Data Warehouse (DW)
- Web and Social Network Service (SNS)
 - Google keyword search, Facebook, Twitter, ...
- Blockchain systems
 - (world) state: managed by key-value DBMSs (e.g., LevelDB, CouchDB)
 - ledger: stored in file systems, but ...
- Data lake
 - unstructured data is processed and fed into OLAP systems
- Retrieval-Augmented Generation(RAG) for Gen AI(LLM)
 - images, videos, texts, time-series (e.g., voice) are stored in vector DB

Relational database systems

- Concept was proposed by Ted Codd in 1970
 - database is organized as a set of tables called relations
 - a relation is composed of a set of tuples (records)
 - all tuples of the same relation have the same set of columns
- Programmers have no concern with the storage structure
- Queries are expressed in a high-level language (SQL)
 - greatly increase the efficiency of database programmers
 - query optimizer must generate an efficient query plan
- Object-oriented features are combined into relation model
 - Object Relational DBMS (ORDBMS) : Oracle, ...

NoSQL database systems (1)

- Latest attempt to overthrow the relation model in 2010s
- Retroactively reinterpreted as Not Only SQL
- Several key factors
 - need for greater scalability (very large datasets, very high write throughput)
 - strong preference for free/open source software over commercial database products
 - specialized query operations that are not well supported by relational model
 - restrictiveness of relational schema (desire for more dynamic and expressive data model)

NoSQL database systems (2)

- Key-value database systems
 - Dynamo, MemcacheDB, Redis, LevelDB, RocksDB, ...
- Document-oriented database systems
 - MongoDB, CouchDB, Riak, ...
- Column-oriented database systems
 - BigTable, HBase, Apache Cassandra, ...

JSON

flexibility

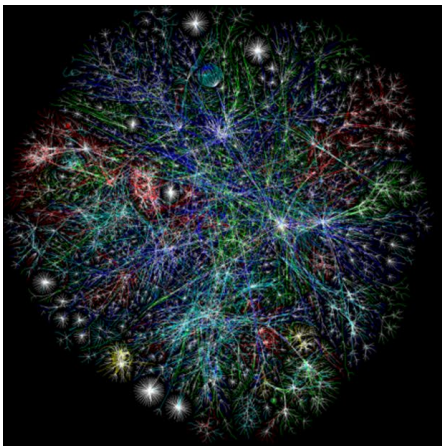
efficiency

join

normalization

Graph database systems

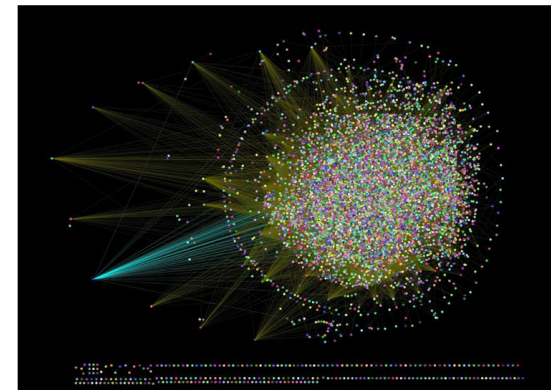
- Graph is one of the most powerful data models
 - consisting of vertices and edges
 - e.g., web, social networks, telecommunication, biology, neuroscience, machine learning, robotics, ...
- Graph DBMSs or frameworks : Neo4j, Tigergraph, Kuzu, ...
 - algorithms: PageRank, connected components, triangle counting
 - queries: Cypher, SQL/PGQ



web



social networks



Protein-Protein-Interaction network

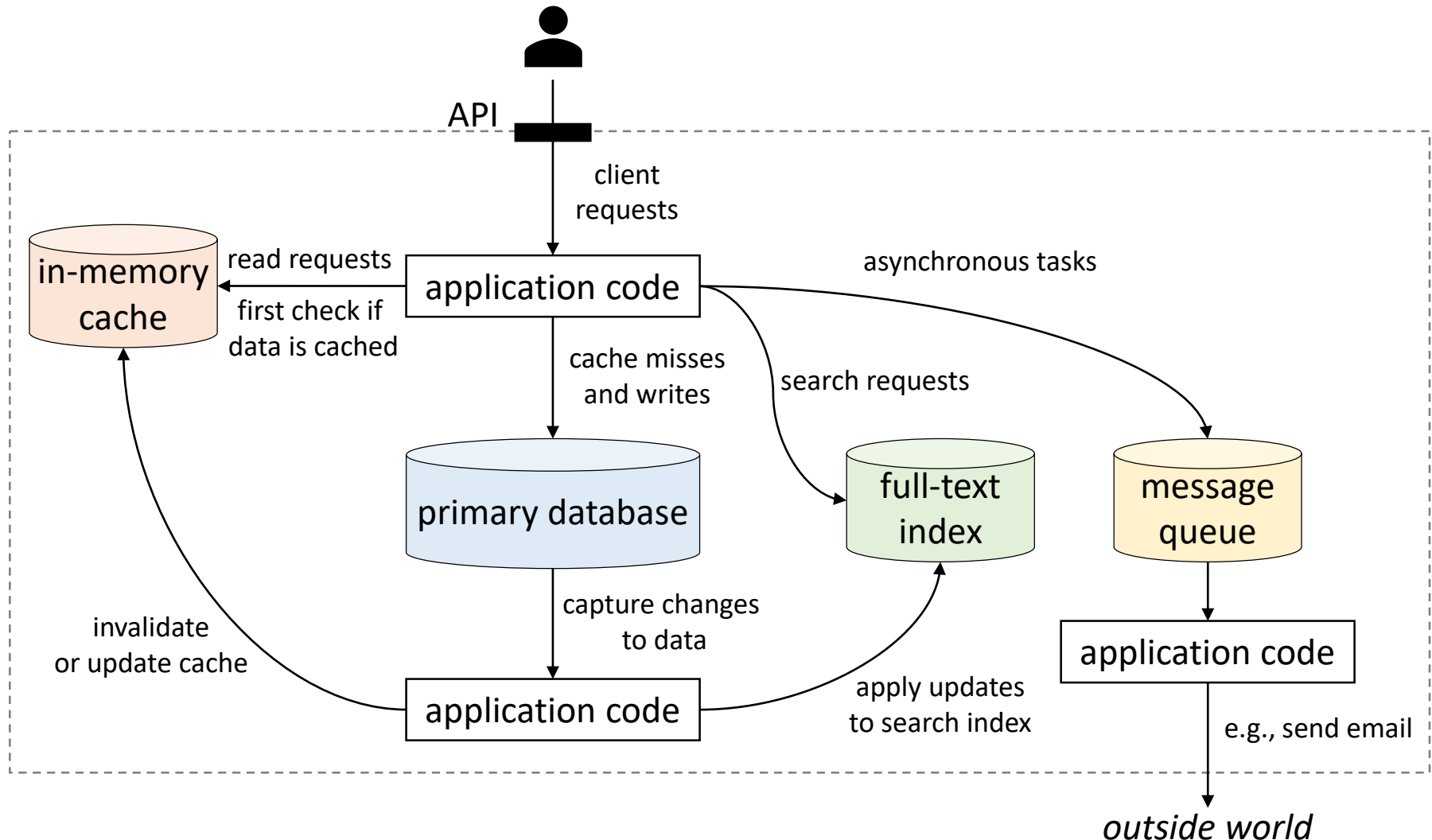
Comparison among database models

Data models	Performance	Scalability	Flexibility	Complexity	Functionality
Relational	variable	variable	low	moderate	relational algebra
Key-value	high	high	high	none	variable (none)
Document-oriented	high	variable (high)	high	low	variable (low)
Column-oriented	high	high	moderate	low	minimal
Graph	variable	variable	high	high	graph theory

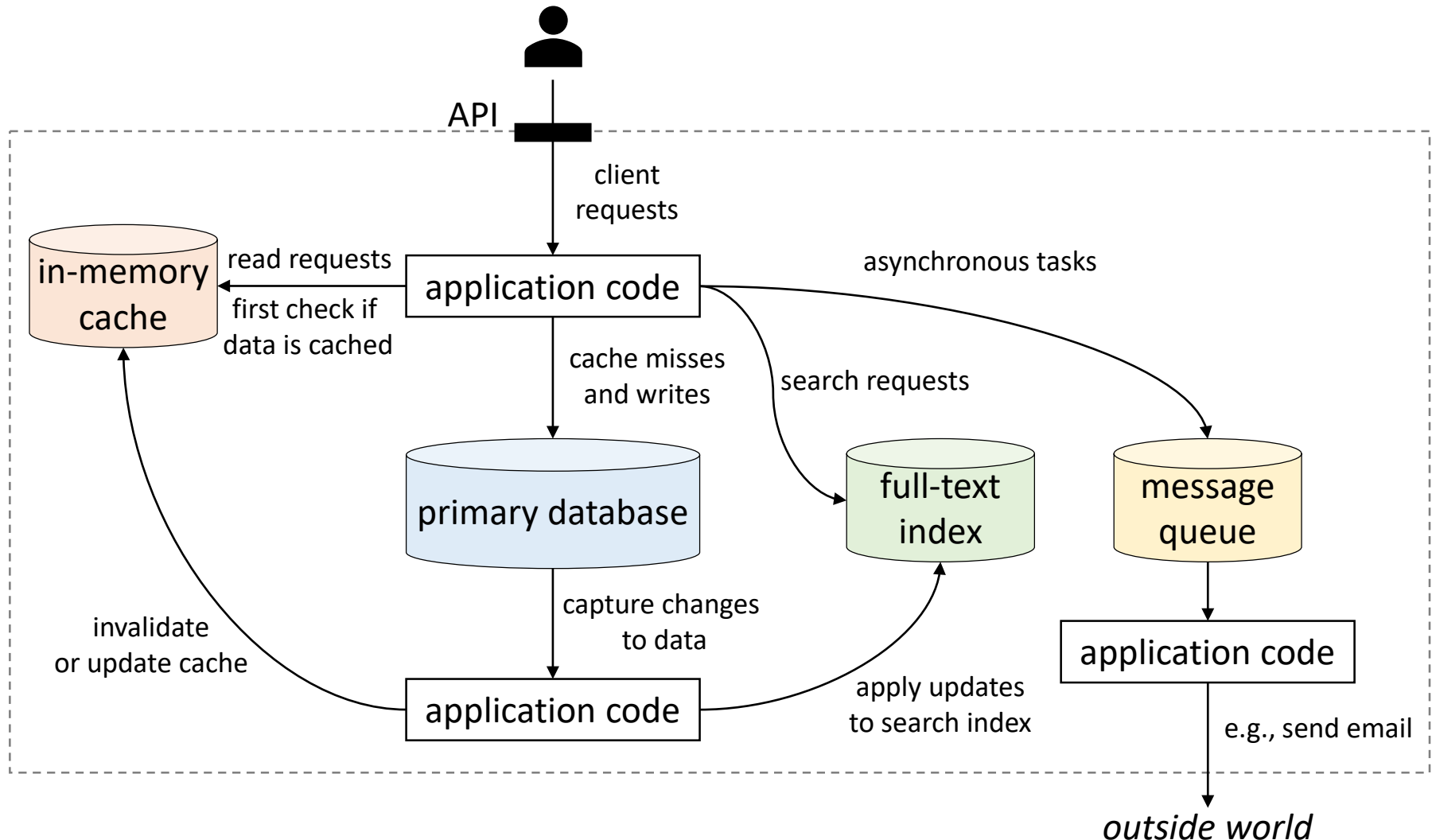
No single perfect database system

- DBMS was the only single data system (past)
- Now, we use a data system combining several components
 - store data so that they can find it again later (database systems)
 - remember the result of an expensive operation (caches)
 - allow users to search data by keyword (search indexes)
 - send a message to another process (stream processing)
 - periodically crunch accumulated data (batch processing)
- Many data systems with different characteristics
 - different applications have different requirements
 - need to figure out which tools and which approaches are the most appropriate for the task at hand

Data system combining several components

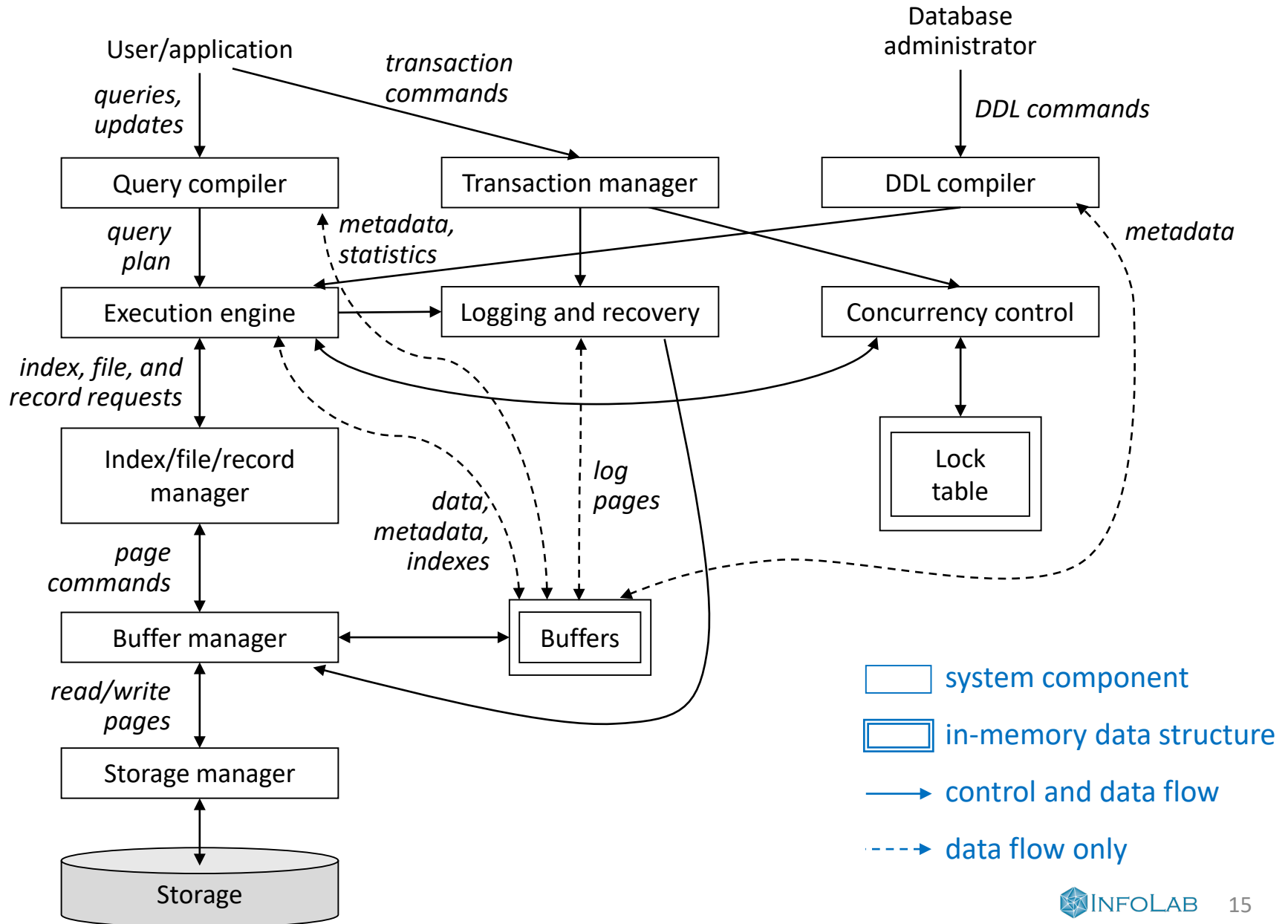


Data system combining several components



Nano quiz (3) (test)

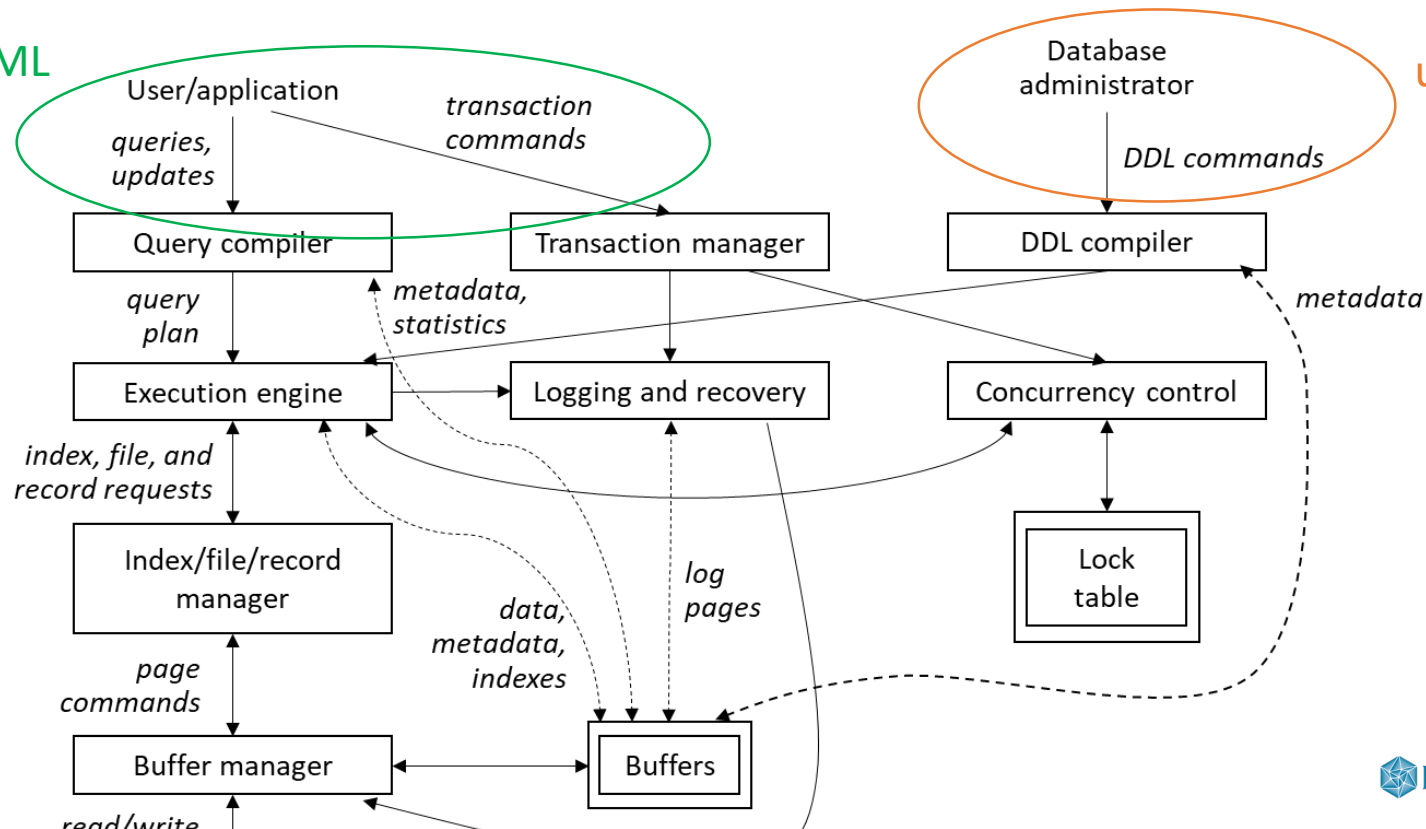
Overview of a DBMS



Two distinct sources of commands to the DBMS

- Conventional users and application programs that ask for data or modify data
- A database administrator (DBA): a person or persons responsible for the structure or schema of the database

using DML

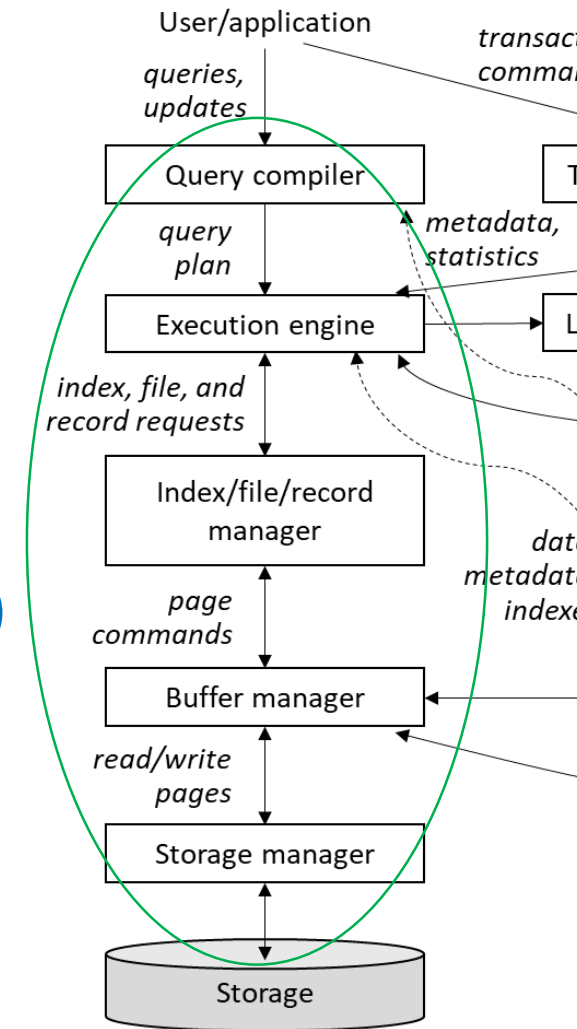


Data-definition language (DDL) commands

- DBA for a university academic affairs database
 - might decide that there should be a table with columns for
 - a student
 - a course that student has taken
 - a grade for that student in that course
 - might also decide that the only allowable grades are
 - A, B, C, D, and F
- **Schema** of database (also called **metadata** of database)
 - includes such a structure and constraint information
 - is defined by DBA's DDL commands

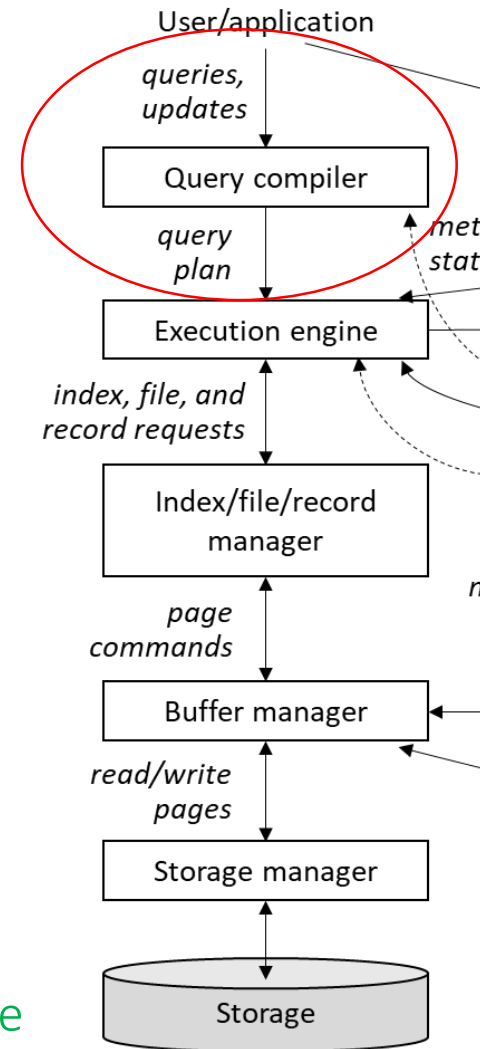
Overview of query processing

- User or application initiates some action, using DML
- **DML: data-manipulation language**
 - not affect the schema of database
 - usually extract data from database (query)
 - may affect the content of database (update)
- DML statements are handled by two separate subsystems
 - query processing (read-only)
 - transaction processing (read-write)



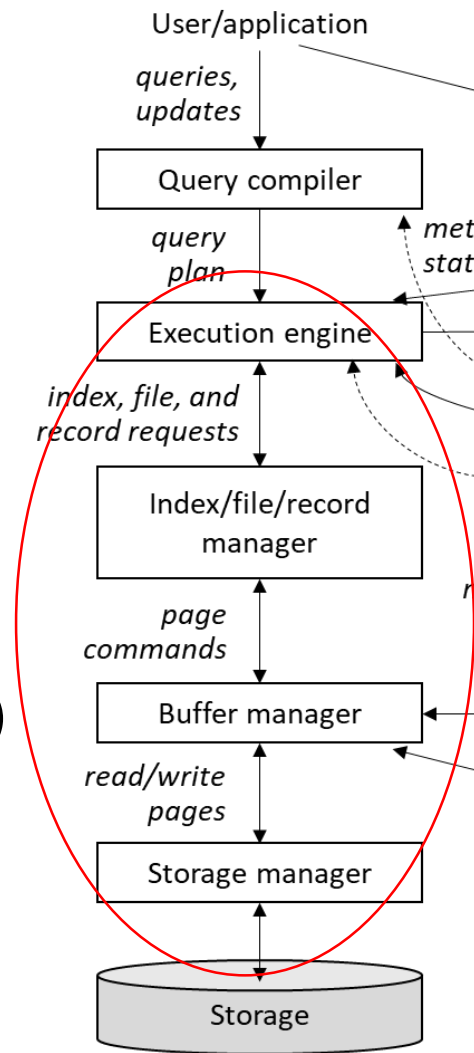
Answering the query (read-only) (1)

- **Query compiler**: translate SQL query into a query plan
 - operations in a query plan are represented in **relational algebra** operations
 - e.g., selection, projection, product, union, ...
- Three major units
 - **query parser**: build a parse tree from query text
 - **query preprocessor**
 - perform semantic checks
 - turn the parse tree into an initial query plan
 - **query optimizer**
 - transform the initial query plan into the best available query plan



Answering the query (read-only) (2)

- The resulting query plan is passed to **execution engine**
- Execution engine issues a seq. of requests
 - for small pieces of data, typically tuples of a relation
 - to a **resource manager** that knows about
 - data files (holding relations)
 - format and size of tuples in those files
 - index files (to help find elements of data files quickly)
- The requests are passed to **buffer manager**
 - to bring appropriate portions of data from disk
- Buffer manager communicates with **storage manager**

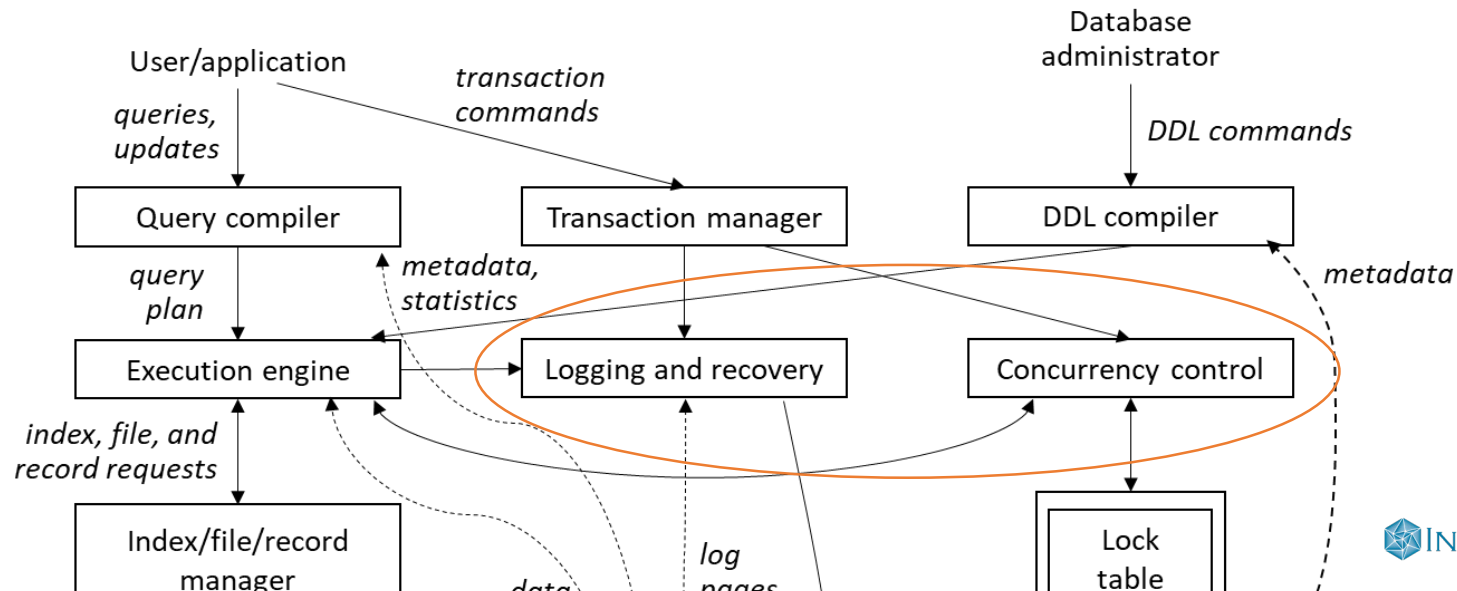


Storage and buffer management

- Data of a database normally resides in disk (HDD or SSD)
- Data must be in main memory to perform any op. on data
- Storage manager
 - control the placement of data on disk and its movement
 - might be the file system of the underlying OS in a simple DBMS
 - but, DBMS's normally control storage on disk directly
- Buffer manager
 - partition the available main memory into buffers
 - transfer disk blocks into page-sized regions (i.e., buffers)

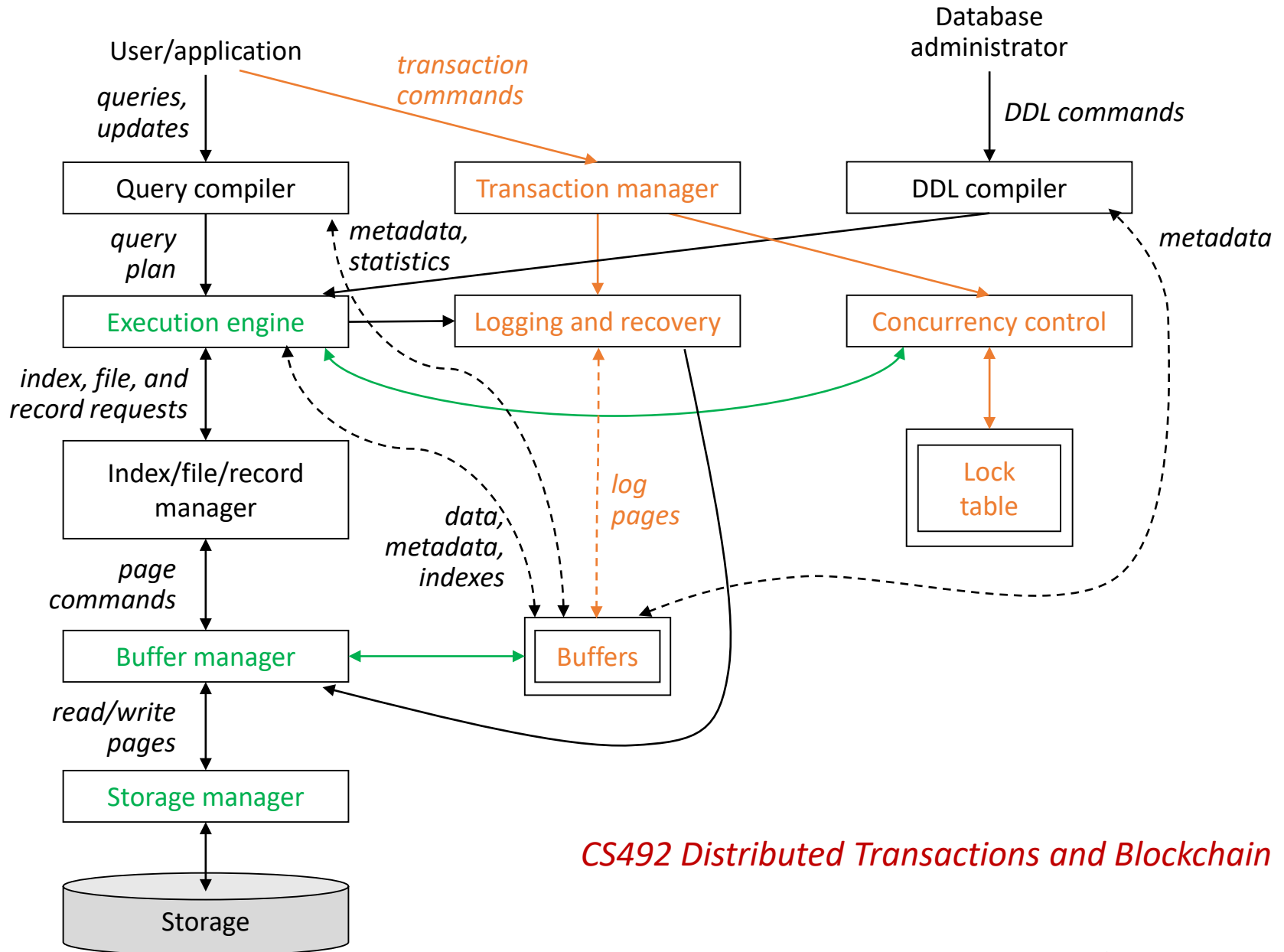
Transaction processing (read-write)

- **Transaction** (Tx): a unit that
 - must be executed **atomically**
 - must be in **isolation** from another Tx
 - must be **durable**
- } **ACID** properties of Tx
(C: consistency)
- effect of any completed Tx must be preserved even if the system fails in some way right after completion of the Tx
- Transaction processor is divided into two parts



- Log manager
 - for durability, every change in DB is **logged** separately on disk
 - initially write the log in buffers (buffers are written to disk)
- Recovery manager
 - no matter when a system failure or crash occurs
 - examine the log of changes and restore DB to **consistent state**
- Concurrency-control manager
 - individual actions of multiple Txs are executed in such an order
 - that the net effect is the same as if the Txs had executed one-at-a-time
 - maintain **locks** on certain pieces of database (e.g., tuples)
 - locks are generally stored in a main-memory lock table
- **Deadlock** resolution
 - Txs compete for resources (e.g., tuples) through locks
 - **transaction manager** cancels (i.e., abort) one or more Txs

Nano quiz (4) (test)



CS492 Distributed Transactions and Blockchain

Outline of this course – Part 1

- Relational model
- The design theory for relational databases
 - functional dependencies, normalization
- High-level database models
 - Entity-Relationship(ER) model
- Algebraic and logical query languages
- Database query language SQL
- Views and indexes
 - e.g., machine for graduation certificate, Bank's ATM, ...

Outline of this course – Part 2

- Secondary storage management
 - how storage can be organized to allow efficient access to data
- Index structures
 - B-tree, multidimensional index, bitmap index, ...
- Query execution
 - algorithms to implement the operations in a query plan
- Query compiler
 - how to select an efficient query plan among all possible ways
- On-line analytic processing (OLAP)
 - data warehouses, OLAP cube, BI, decision support, ...

Thank you

Q & A